

Quantum-Classical Equivalence for AND Functions

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Quantum-Equivalence Conjecture: For a total function $f : \{0, 1\}^n \rightarrow \{0, 1\}$,

$$R(f) \leq \text{poly}(C(f))$$

where

- $R(f)$ = classical randomized communication complexity
- $Q(f)$ = quantum communication complexity (with prior entanglement)